

THE DIGITAL TWIN IS NOT THE AUTHORITY

Atmospheric Integrity Records, Admissible Execution Architecture,
and the independent governance layer between physical reality
and authorized consequence

Governance does not need to become the technology it governs.

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Independent architectural demonstration. Not a NIST or ASHRAE endorsement, certification, implementation claim,
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THE GOVERNANCE OVERLAY

The one-minute architectural explanation

The building, its instruments, its models, and its execution systems retain their existing responsibilities. TA14 does not replace them. It establishes examinable evidence and authority boundaries across them.

Existing operational systems	Independent governance responsibilities
Physical environment + sensors: observe conditions	AIR: preserve the evidence basis, provenance, continuity, uncertainty and raw observations
Digital twin + analytics: model, infer, predict, recommend	Distinguish observed measurements from model assertions and proposed consequences
Owner + authorized controllers: decide and execute locally	AEA: examine evidence, authority and standing for a bounded proposed consequence
Post-action measurement systems: observe what happened	Preserve execution receipts and independently examine proof of performance

Three boundaries that must never collapse

Observation is not inference. A model-generated estimate cannot silently replace missing measurements.

Inference is not authorization. A technically sound recommendation is not permission to change an asset.

Execution is not proof. A completed command cannot stand in for a verified physical outcome.

Governing question: Does this proposed consequence have sufficient Admissible Evidence, Applicable Authority, and Established Standing to become reality NOW?

Conceptual reference architecture only. No deployed integration or third-party endorsement is claimed.

Executive proposition

Digital twins, building automation, environmental sensors, and AI systems can observe, represent, simulate, recommend, and sometimes execute changes in physical environments. Those capabilities do not independently establish that an observation is admissible evidence or that a proposed intervention is authorized.

TA14 separates these responsibilities. Atmospheric Integrity Records (AIR) establish requirements for preserving and examining atmospheric evidence without replacing the instruments or platforms that produce and store it. Admissible Execution Architecture (AEA) examines the boundary between a proposed consequence and real-world execution without becoming the AI, digital twin, controller, or actuator.

Does this proposed consequence have sufficient Admissible Evidence, Applicable Authority, and Established Standing to become reality NOW?

The opportunity for digital-twin developers is an independent, technology-neutral governance overlay. The opportunity for building owners is a more explicit basis for examining evidence, authority, intervention, and outcome. The opportunity for research is to test whether these boundaries can be expressed as interoperable, auditable interfaces.

What this reference does - and does not - establish

This reference describes a proposed architecture and a testable demonstration pathway. It does not assert that a NIST system has integrated TA14, that an AIR automatically validates a sensor, that a model recommendation grants permission, or that the proposed commercial model has been adopted by the market.

01 Physical reality precedes its representation

A building exists and behaves independently of its dashboards, models, and records. Indoor pressure, airflow, temperature, particulate concentrations, occupancy, equipment state, and environmental exposures are physical conditions. A digital representation can approximate these conditions; it is not identical to them.

A hospital digital twin may display an operating-room pressure relationship or estimate how a ventilation change will affect adjacent spaces. Its representation is only as useful as the evidence, model assumptions, time alignment, and operating context supporting it.



The architectural rule is straightforward: observations must remain distinguishable from derived calculations, assumptions, predictions, and simulated states. If a model interpolates a missing sensor interval, that inferred value must not silently replace the original absence of evidence.

02 AIR is an evidentiary governance architecture

An Atmospheric Integrity Record is not a new type of sensor and does not require ownership of a database, dashboard, or digital-twin application. It specifies the evidentiary context needed to preserve, examine, and compare atmospheric observations.

Minimum examinable record dimensions

Dimension	What should remain examinable
Observation	Measured quantity, units, timestamp, source, original value, and acquisition context.
Instrument	Identity, location, method, calibration status or available evidence, and limitations.
Operating context	Equipment state, occupancy or relevant building conditions, and known external influences.
Continuity	Gaps, delays, substitutions, changes, version history, and chain of custody where applicable.
Uncertainty	Measurement uncertainty, missing context, confidence limits, and inference labels.
Comparison	Preserved pre-intervention baseline and post-intervention records, with adjustments separated from raw observations.

A governed record can expose insufficiency rather than conceal it. A missing calibration certificate, an interrupted chronology, or uncertain instrument placement may limit what the evidence can support. AIR does not transform such limitations into certainty.

03 The digital twin interprets; it does not confer authority

Digital twins may assimilate sensor data, engineering models, equipment metadata, simulation outputs, and predictions. They can be useful for identifying deviations, evaluating scenarios, and proposing interventions. But interpretation is a separate function from preserving evidence, and neither is equivalent to authorization.

A recommendation to increase outdoor airflow could be technically reasonable while still conflicting with a pressure-control requirement, an infection-control constraint, an energy limit, a maintenance lockout, or the building owner's operating authority. The model may propose; it does not thereby receive permission.

Observed is not inferred. Inferred is not authorized.

A practical integration should preserve the identity of the evidence records used, the version of the model or rule set, the assumptions applied, the proposed action, and the declared limits of the recommendation. The governance interface should not need to inspect every internal algorithm to identify the action and its supporting record references.

04 The proposed consequence is a separate object

Before an intervention is evaluated, the recommendation should be expressed as a specific proposed consequence: the target asset, requested change, time window, operating bounds, requesting actor, claimed purpose, evidence references, and expected result.

“Improve indoor air quality” is a goal, not a sufficiently bounded execution request. “Increase outside-air damper minimum from X to Y on air-handling unit Z for a defined interval, subject to pressure and safety limits” is closer to an examinable proposal. The exact parameters must be established by competent authorized parties; the values here are deliberately unspecified.

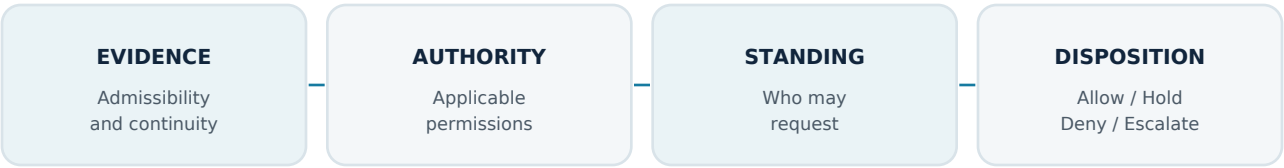
Proposed-consequence envelope

Field	Example of required specificity
Asset / scope	Which building, air handler, zone, or system is affected?
Requested action	What setting or physical state is proposed to change?
Temporal validity	When does the request apply, and when does it expire?
Evidence basis	Which preserved observations and model outputs support it?
Constraints	What operating, safety, clinical, or owner-defined limits apply?
Requesting standing	Who is asking, in what capacity, and under which authority chain?

05 AEA examines the authority boundary

Admissible Execution Architecture (AEA) does not replace the digital twin, the AI agent, the building automation system, or the responsible human decision-maker. It addresses the separate question of whether the proposed consequence has sufficient evidence, applicable authority, and established standing to proceed now.

The examination should distinguish evidence sufficiency from authority sufficiency. A valid sensor record does not confer authority. A properly authorized operator does not make weak evidence stronger. A previous authorization does not necessarily survive a material change in context.



Four dispositions

Disposition	Meaning at the proposed boundary
ALLOW	Specified conditions are satisfied for this bounded proposed consequence; downstream execution remains subject to local controls.
HOLD	Required evidence, context, or conditions are incomplete; preserve the boundary while resolving the deficiency.
DENY	The proposed consequence fails an applicable requirement or lacks authority under the established rules.
ESCALATE	The matter requires another designated decision-maker, examination, or authority process.

The disposition must be tied to the actual proposal, supporting records, applicable rules, and valid time context. It is not a blanket permission to operate the building.

06 Execution authority must be established locally

The digital twin may reside in a vendor cloud, the AIR record service in another environment, and the building automation controller at the facility. Context can travel across these systems. Execution authority cannot simply be assumed to travel with it.

TA14's federation principle is that authority context may cross boundaries while execution authority must be established locally. The recipient must determine whether the proposed action is permitted for the particular asset, time, conditions, and governing authority.

The crossing is not the permission. Capability is not authority. Commissioned once is not authorized forever.

An ALLOW disposition from a governance examination is not a command to bypass a safety interlock, owner approval, emergency procedure, or applicable regulatory requirement. Existing authorized systems remain responsible for performing the action within their own enforceable constraints.

07 Execution is not proof of outcome

A command accepted by a controller is not the same as a successful physical intervention. An actuator moving is not the same as the intended environmental result. A predicted improvement is not a verified improvement.

After an authorized action, the system must preserve what was actually executed, when it occurred, whether it completed or failed, and what subsequent measurements demonstrate. The post-performance record should remain distinct from the proof-of-performance examination.

Evidence-to-outcome comparison

Stage	Question
Original state	What did the building demonstrably look like before intervention?
Authorized action	What exactly was approved, by whom, and under which bounded conditions?
Execution record	What command or human intervention actually occurred?
Post-performance record	What did the building subsequently do, according to preserved observations?
Proof of performance	What comparison and uncertainty assessment support the outcome claim?

An outcome may be improved, restored, unchanged, degraded, or unresolved. A governance system must be capable of recording unfavorable or inconclusive results rather than treating execution as automatic success.

08 One independent overlay across many technologies

The central design opportunity is independence. TA14 need not build a competing digital twin, replace environmental monitoring, or become the control platform. Instead, implementers can expose evidence references, proposed consequences, authority context, dispositions, execution receipts, and outcome records through governed interfaces.



WORKED EXAMINATION

One bounded hospital ventilation proposal - illustrative, not operational

A digital twin detects an apparent deterioration in air conditions and proposes adjusting an air-handling unit. The scenario intentionally contains no numeric setpoint: those values belong to authorized site personnel and applicable constraints.

Artifact	Examinable question / content
AIR evidence	Instrument ID, timestamp, raw reading, operating state, calibration evidence, missing intervals, uncertainty
Twin assertion	Model/version, observed inputs vs inferred states, prediction, assumptions and known limits
Proposed consequence	Specific asset and proposed adjustment, time window, requesting actor, safety bounds and expected result
Authority context	Who can authorize this action? What scope, revocation, time and local restrictions apply?
Execution and outcome	Local control check, execution receipt, new raw observations, baseline comparison and unresolved uncertainty

Four possible dispositions

Result	Illustrative condition
HOLD	Critical pressure-safety evidence is missing or the atmospheric record has a material continuity gap.
DENY	The requester lacks authority under the applicable established rules.
ESCALATE	The proposed action requires clinical or designated owner review beyond the requester's standing.
ALLOW	Evidence, authority and standing are sufficient for this specific bounded request; local safety controls still govern execution.

An ALLOW is not a universal building operating permission. A post-action comparison may still find the outcome unchanged, degraded, or unresolved.

This independence is what makes the architecture potentially reusable across schools, hospitals, commercial buildings, industrial sites, infrastructure, and environmental programs. The underlying technologies differ. The governance questions remain recognizable.

It also establishes an important institutional boundary: the governance institution can define and examine the method without claiming ownership of the assets, physical measurements, model algorithms, or execution controls.

A concrete digital-twin examination scenario

Illustrative hospital ventilation case - not an operational pilot

1. Observe. The building monitoring system reports a possible deterioration in indoor atmospheric conditions. Relevant raw observations and operating conditions are preserved.
2. Establish record continuity. AIR requirements are applied to examine instrument identity, timestamps, missing intervals, calibration evidence, and the original operating baseline. Unknowns remain explicit.
3. Interpret. The digital twin analyzes the available evidence and proposes a bounded ventilation adjustment. The model's prediction is labeled as a prediction, not an observation.
4. Examine the consequence. AEA evaluates the particular request against admissible evidence, applicable authority, and established standing. An unresolved pressure-safety constraint, for example, could require HOLD or ESCALATE rather than execution.
5. Execute locally if authorized. The facility's authorized control or human operating process independently enforces the applicable local conditions. No external model gains authority merely by sending a message.
6. Verify. New observations and operating records are preserved. The post-performance evidence is compared with the original baseline, with uncertainty and alternative explanations documented.

This scenario is suitable for a technical demonstration because it tests each boundary separately. It should begin with simulated or otherwise safely isolated controls, not an unreviewed live-building intervention.

Proposed interoperability contract for examination

The following is a candidate interface structure for discussion, not a published interoperability standard or deployed product specification.

Exchange artifact	Required content
Evidence reference	Immutable or versioned identifier; source; timestamps; provenance; continuity and uncertainty status.
Model assertion	Model identity/version; inference or prediction; supporting evidence; assumptions; confidence or limits.
Consequence proposal	Target; requested action; bounds; validity period; requester; expected result; relevant constraints.
Authority context	Local authority references; standing; delegated scope; conditions; expiration; revocation state.
Governance disposition	ALLOW / HOLD / DENY / ESCALATE; rationale; ruleset version; examined inputs; validity and constraints.
Execution receipt	Local authorization check; command or intervention record; executor; time; success/failure status.
Outcome evidence	Post-action observations; raw-versus-adjusted distinction; comparison method; uncertainty; result.

Non-negotiable design properties

A credible pilot should demonstrate that missing evidence cannot silently become admissible, model output cannot silently become observation, a transmitted disposition cannot override local authority, stale approvals cannot silently persist, and an execution receipt cannot silently become proof of performance.

Commercialization: what could actually be offered

A technology-neutral overlay suggests several potential service and product forms. These are commercialization hypotheses to validate through customer discovery and controlled demonstrations; they are not claims of market acceptance.

Potential offer	Buyer-facing value / test
Technical pre-examination	Assess a digital-twin workflow against evidence, authority, and outcome boundaries; produce an explicit gap report.
Registered governance pathway	Establish a versioned governance baseline and a defined scope for subsequent examinations.

CANDIDATE INTERFACE

What a digital-twin integrator would submit and receive

This is a conceptual exchange envelope for a technical pressure test. It is not a published standard, API contract, certification, or working integration.

Input / output	Minimum meaning
Evidence reference (input)	Versioned ID, observation provenance, continuity and uncertainty status; original observations retained
Model assertion (input)	Model version, prediction or inference label, evidence references, assumptions and confidence limits
Consequence proposal (input)	Target asset, requested action, bounds, validity period, requesting actor and expected result
Authority context (input)	Applicable local authority, delegation and standing, constraints, expiry and revocation state
Disposition (output)	ALLOW / HOLD / DENY / ESCALATE; rationale, ruleset version, examined inputs, validity and constraints
Execution receipt (later)	Local authorization check, executor, action actually attempted, time and success/failure
Outcome evidence (later)	Preserved post-action observations, comparison method, uncertainty and supported result

Minimum pressure tests

1. Missing observations stay missing, even if a model interpolates them.
2. Expired or revoked authority is rejected at the local execution boundary.
3. A disposition cannot bypass a local safety interlock or owner requirement.
4. A successful execution receipt cannot be promoted automatically to proof of performance.
5. Raw and adjusted observations remain separately identifiable with method and uncertainty.

Potential offer	Buyer-facing value / test
Interoperability profile	Specify a bounded exchange of evidence, proposals, dispositions, and receipts across vendors.
Pilot / reference implementation	Demonstrate a simulated or approved building scenario with auditable decision and outcome records.
Operational governance service	Periodically examine changes in authority context, evidence continuity, and execution conditions.

Commercial terms, liability allocation, integration costs, legal applicability, and the actual willingness of buyers to pay must be established separately. A technical architecture is not, by itself, a procurement commitment or certification program.

A practical research invitation

A useful first collaboration would be a bounded technical pressure test, not a request for institutional endorsement. Select one digital-twin ventilation recommendation, preserve a small but realistic atmospheric evidence set, define an explicit proposed consequence, and test whether independent authority examination changes the decision path.

Questions for researchers and developers

Can the digital twin distinguish observed inputs from inferred states? Can it carry AIR evidence references without owning the AIR architecture? Can a proposed action be expressed as a bounded consequence? Can a local system independently validate authority? Can an expired or revoked authorization be rejected? Can post-intervention records demonstrate an unresolved or negative outcome without being overwritten by the model's prediction?

A successful examination would not prove universal applicability. It would demonstrate that the architectural boundaries can be represented, tested, and challenged in one clearly specified setting.

Conclusion: govern the crossing, not the technology

The industry does not need another platform claiming to replace every existing system. It needs a disciplined way to preserve evidence, distinguish interpretation from observation, establish authority, constrain execution, and verify consequence.

The digital twin may represent reality. Atmospheric Integrity Records govern the evidentiary integrity of environmental observations. Admissible Execution Architecture examines whether a

FROM EXAMINATION TO ENGAGEMENT

A bounded, buyer-facing pathway

The commercial proposition is not that TA14 owns the digital twin. It is that a buyer may benefit from an independently examinable governance boundary that can operate alongside existing technology.

Stage	Deliverable	What is not implied
01 - Discover	Identify one real decision boundary and the responsible asset owner	No claim of procurement or product-market fit
02 - Register governance	Establish steward, scope, version and frozen baseline before a formal challenge or examination	Registration is not endorsement or automatic approval
03 - Founding demonstration	One bounded, safely simulated examination of evidence, standing, disposition and outcome	No live control access or unreviewed site intervention
04 - Examine / establish	Document gaps, authority dependencies and an agreed interoperability profile	No automatic regulatory or legal certification
05 - Operate (if contracted)	Versioned evidence, authority and revalidation practices with clear responsibilities	No perpetual authorization or guaranteed outcome

A useful commercial discovery question

Which proposed physical consequence in your digital-twin workflow is currently difficult to justify, authorize, or verify - and what would an independent, auditable decision record be worth to its owner?

Any pricing, liability allocation, intellectual-property license, integration cost, security requirements and service obligations require a separate agreement. No market validation or existing sale is represented.

RESEARCH INVITATION

A bounded technical examination suitable for independent scrutiny

Proposed study: one safely simulated hospital ventilation recommendation, one preserved atmospheric evidence set, one explicitly bounded proposed consequence, and one local authority boundary.

Phase	Observable success criterion
Baseline	Raw environmental observations, instrument context, time alignment and gaps are preserved and distinguishable from model outputs.
Proposal	The twin expresses a specific action, target, duration, assumptions and constraints without claiming execution permission.
Examination	Evidence, applicable authority and established standing produce a traceable ALLOW, HOLD, DENY or ESCALATE decision.
Local enforcement	A local authority or safety constraint can reject the proposed action even when the external recommendation is technically plausible.
Outcome	Execution receipt and post-performance evidence remain separate; inconclusive or unfavorable results are reportable.

Questions to invite, not conclusions to assert

Can the existing digital-twin stack reference AIR evidence without absorbing AIR? Can the authority boundary remain independent of the model? Can changed conditions trigger revalidation? Can a failed or uncertain outcome remain visible rather than being rewritten as success?

Invitation: Researchers, building owners, and digital-twin developers are invited to challenge the proposed boundaries using a narrowly specified example. This document does not claim NIST, ASHRAE, hospital, or vendor endorsement.

Institutional reference: <https://www.ta14exchange.com/>

proposed consequence may cross into authorized execution. Existing systems carry out the permitted action. New records support examination of what actually happened.

The digital twin is not the authority. The sensor is not the evidence standard. The command is not the outcome. Governance is the boundary between knowing, proposing, and making real.

Institutional scope and attribution

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Limitations: No NIST, ASHRAE, hospital, digital-twin provider, or other institutional endorsement is claimed. No operational integration, certification, formal examination disposition, or performance guarantee is represented by this publication. Examples are illustrative and require site-specific professional, safety, legal, and owner-authority review before any real-world use.

Reference architecture: Reality → Record → Continuity → Admissibility → Binding → Commit → Execution → Outcome. The framework is intended to preserve the separation between observation, inference, admissibility, authority, execution, and outcome.